INCH - POUND

MS20028G
31 August 2010
SUPERSEDING
MS20028F
20 May 2005

DETAIL SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, STRAIGHT, SOLDER TYPE, BAYONET COUPLING, CLASS T, SERIES I

Inactive for new design after 7 December 1998.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27599.

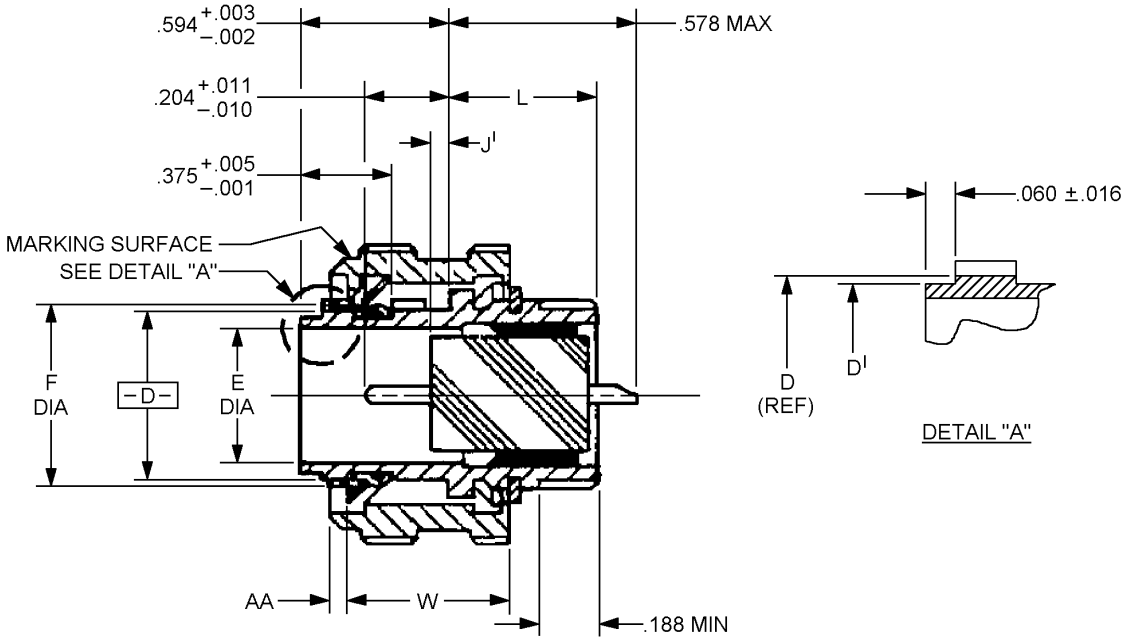
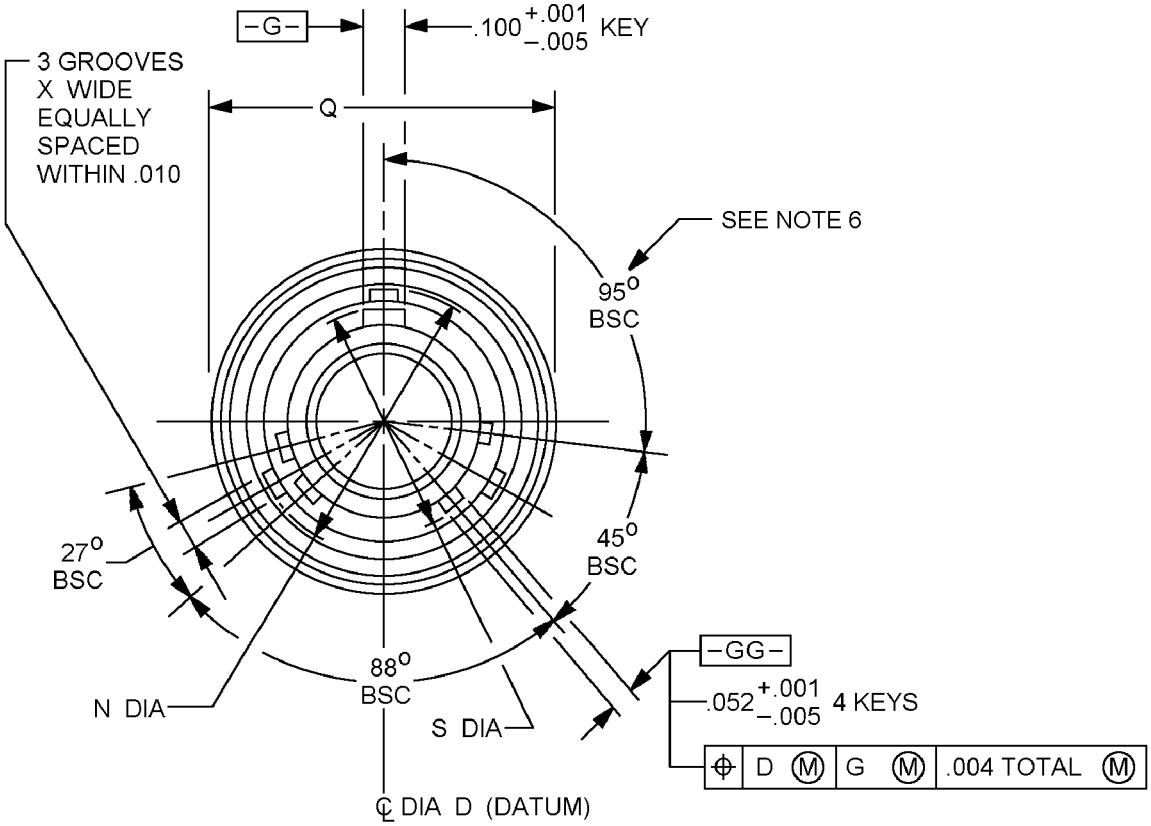


FIGURE 1. Plug.

AMSC N/A

FSC 5935

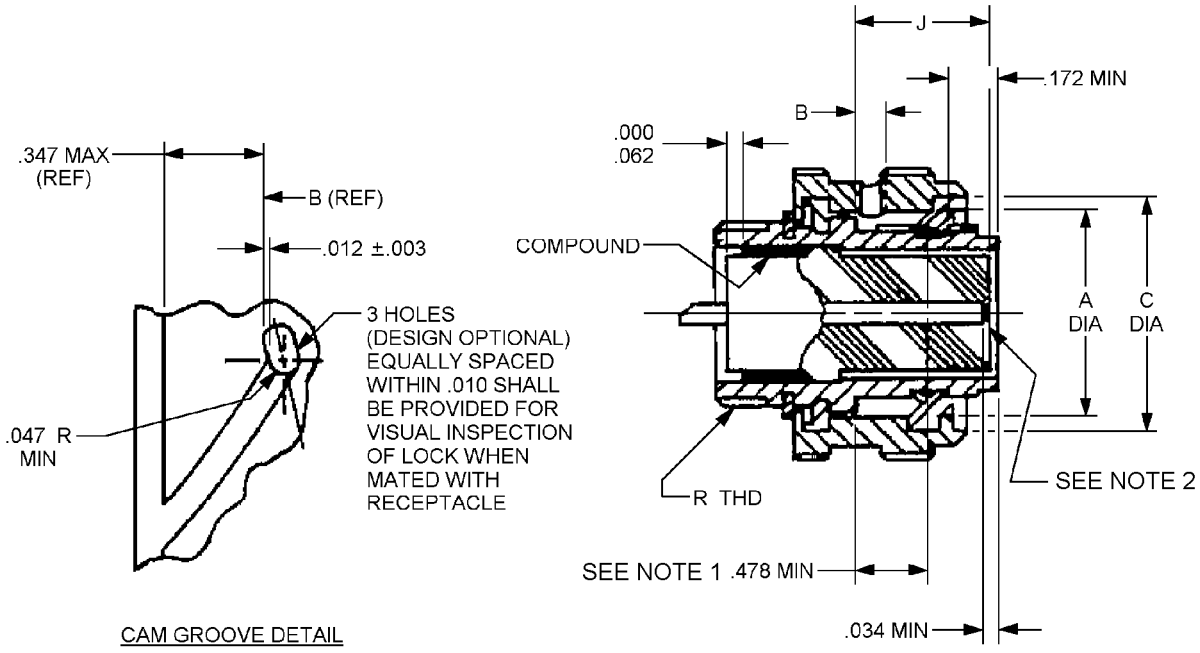
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CLASS T

FIGURE 1. Plug - Continued.

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NOTES:

1. The point at which a gauge pin having the same basic diameter as the mating contact and a square face, touches socket contact spring.
2. Insert front surface shall be flat within .005 T.I.R.
3. Diameter D with respect to diameter S shall be concentric at MMC.
4. Diameters A and N shall be concentric within .008 T.I.R.
5. Dimensions are in inches. Metric equivalents are given for information only.
6. Normal keyway position. For other keyway positions, see MIL-DTL-27599.
7. Finish: A - Bright cadmium (see MIL-DTL-27599).
   B - Olive drab (see MIL-DTL-27599).
   F - Electroless nickel in accordance with SAE-AMS-C-26074, class 3 or 4, grade B, finish dull.
   T - Nickel flurocarbon (see MIL-DTL-27599).
   Z - Zinc nickel (see MIL-DTL-27599).

FIGURE 1. Plug - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.001	0.0254	0.011	0.280	0.06	1.524	0.347	8.814
0.002	0.051	0.012	0.305	0.062	1.575	0.375	9.525
0.003	0.076	0.016	0.406	0.100	2.54	0.478	12.14
0.004	0.102	0.034	0.864	0.172	4.370	0.578	14.68
0.005	0.127	0.047	1.194	0.188	4.775	0.594	15.09
0.010	0.254	0.052	1.321	0.204	5.182		

FIGURE 1. Plug - Continued.

TABLE I. Dimensions. 1/ 2/

Shell size	A dia + .005 - .001	B ± .020	C dia + .005 - .001	D dia + .001 - .005	D[I] dia min	E dia ± .003	F dia min
9	.580 (14.73)	.113 (2.87)	.684 (17.37)	.437 (11.10)	.417 (10.60)	.315 (8.00)	.447 (11.35)
11	.712 (18.08)		.832 (21.13)	.565 (14.35)	.545 (13.84)	.433 (11.00)	.575 (14.60)
13	.861 (21.87)		.981 (24.92)	.677 (17.20)	.657 (16.68)	.547 (13.90)	.692 (17.57)
15	.985 (25.02)		1.106 (28.09)	.802 (20.37)	.782 (19.86)	.672 (17.07)	.817 (20.75)
17	1.110 (28.20)		1.230 (31.24)	.927 (23.54)	.907 (23.04)	.797 (20.24)	.942 (23.92)
19	1.217 (30.91)		1.347 (34.21)	1.032 (26.21)	1.012 (25.70)	.886 (22.50)	1.047 (26.60)
21	1.342 (34.08)		1.462 (37.13)	1.157 (29.38)	1.137 (28.88)	1.011 (25.68)	1.172 (29.76)
23	1.467 (37.26)		1.587 (40.31)	1.282 (32.56)	1.262 (32.05)	1.136 (28.85)	1.297 (32.94)
25	1.592 (40.44)	.152 (3.86)	1.712 (43.48)	1.407 (35.73)	1.387 (35.22)	1.260 (32.00)	1.442 (36.62)

See notes at end of table.

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TABLE I. Dimensions - Continued. 1/ 2/

Shell size	J ± .005 socket insert	J[l] ± .005 pin insert	L max	N dia + .005 - .006	Q dia max	S dia + .001 - .010
9	.576 (14.63)	.018 (.457)	.547 (13.90)	.674 (17.12)	.859 (21.82)	.482 (12.24)
11				.822 (20.87)	.984 (25.00)	.610 (15.49)
13				.972 (24.68)	1.156 (29.36)	.759 (19.27)
15				1.097 (27.86)	1.281 (32.53)	.884 (22.45)
17				1.222 (31.03)	1.406 (35.71)	1.009 (25.63)
19				1.329 (33.75)	1.516 (38.50)	1.114 (28.30)
21				1.454 (36.93)	1.641 (41.68)	1.239 (31.47)
23				1.579 (40.10)	1.766 (44.85)	1.364 (34.64)
25			.609 (15.47)	1.704 (43.28)	1.891 (48.03)	1.489 (37.82)

See notes at end of table.

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TABLE I. Dimensions – Continued. 1/ 2/

Shell size	W +.016 -.006	X +.016 -.001	R thread class 2A plated UNEF	AA +.005 -.006
9	.693 (17.60)	.090 (2.29)	7/16 - 28	.068 (1.73)
11	.682 (17.32)		9/16 - 24	
13			11/16 - 24	
15	.693 (17.60)		13/16 - 20	
17			15/16 - 20	
19			1 1/16 - 18	
21			1 3/16 - 18	
23			1 5/16 - 18	
25	.794 (20.16)	.137 (3.47)	1 7/16 - 18	.047 (1.20)

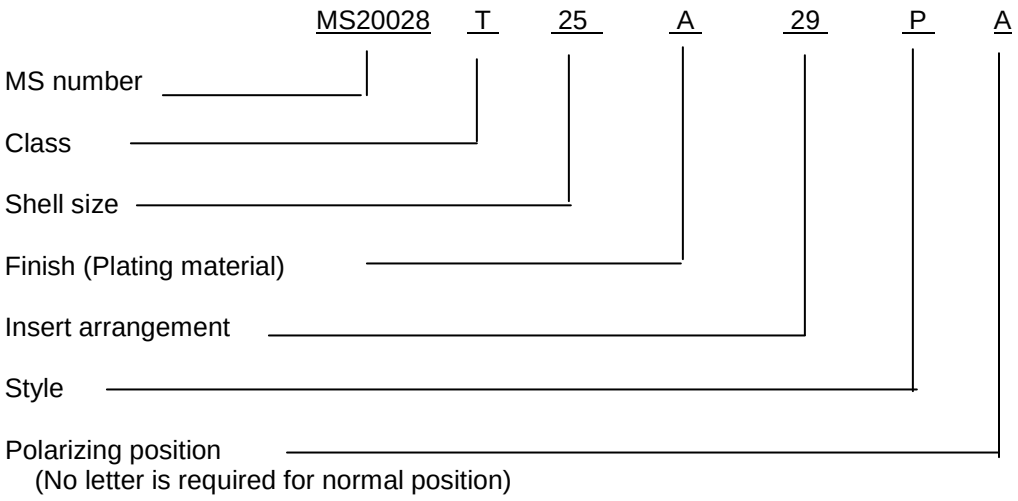
1/ Dimension are in inches.
2/ Metric equivalents are given for information only.

REQUIREMENTS:

Dimensions and configurations: See figure 1 and table I.
This connector mates with MS20026 and MS20029.
Insert arrangement shall be in accordance with MIL-STD-1560.

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Part or Identifying Number (PIN) example:



Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-27599, this document references the following:

MIL-STD-1560
MS20026
MS20029
SAE-AMS-C-26074

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CONCLUDING MATERIAL

Custodians:
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Air Force - 99
DLA – IS

(Project 5935-2010-097)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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